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Course: Calculus Beta

Assignment: MyLab Math - Uge 2

Use implicit differentiation to find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

$$5x^2 + 3y^2 = 16$$

To find $\frac{dy}{dx}$ using implicit differentiation, first differentiate both sides of the equation with respect to x , treating y as a function of x and using the chain rule.

$$5x^2 + 3y^2 = 16$$

$$\frac{d}{dx}(5x^2) + \frac{d}{dx}(3y^2) = \frac{d}{dx}(16)$$

Use the constant multiple rule and the power rule to find $\frac{d}{dx}(5x^2)$.

$$10x + \frac{d}{dx}(3y^2) = \frac{d}{dx}(16)$$

Again use the constant multiple rule and the power rule to find $\frac{d}{dx}(3y^2)$. Treat y as a differentiable function of x .

$$10x + 6y \frac{dy}{dx} = \frac{d}{dx}(16)$$

Note that the right side of the original equation is the constant function, 16, and the derivative of a constant is 0.

$$10x + 6y \frac{dy}{dx} = 0$$

Now solve the equation $10x + 6y \frac{dy}{dx} = 0$ for $\frac{dy}{dx}$.

$$\frac{dy}{dx} = -\frac{5x}{3y}$$

Therefore, $\frac{dy}{dx} = -\frac{5x}{3y}$.

To find $\frac{d^2y}{dx^2}$, differentiate the expression for $\frac{dy}{dx}$ using the quotient rule.

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(-\frac{5x}{3y} \right) \\ &= -\frac{d}{dx} \left(\frac{5x}{3y} \right) && \text{Use the constant multiple rule.} \\ &= -\frac{15y - 5x \left(3 \frac{dy}{dx} \right)}{9y^2} && \text{Use the quotient rule.} \end{aligned}$$

Substitute $-\frac{5x}{3y}$ for $\frac{dy}{dx}$.

$$-\frac{15y - 5x \left(3 \frac{dy}{dx} \right)}{9y^2} = -\frac{15y - 15x \left(-\frac{5x}{3y} \right)}{9y^2}$$

$$= -\frac{15y \cdot y + \frac{25x^2}{y} \cdot y}{9y^2 \cdot y}$$

Multiply the numerator and denominator by y.

$$= -\frac{15y^2 + 25x^2}{9y^3}$$

Simplify.

Therefore, $\frac{d^2y}{dx^2} = -\frac{15y^2 + 25x^2}{9y^3}$.